

Internship Advnce SQL Tasks

1: Add Some Data in the Existing Table

The screenshot shows the MySQL Workbench interface. The 'Query 1' window contains the following SQL code:

```
109
110 • INSERT INTO Employee VALUES
111   (112,'Anjali',68000,'2026-06-10',1),
112   (113,'Rakesh',72000,'2026-06-12',2),
113   (114,'Meena',48000,'2026-06-15',3),
114   (115,'Suresh',90000,'2026-06-18',4),
115   (116,'Kavita',62000,'2026-06-20',2);
116 • SELECT * FROM Employee;
```

The 'Result Grid' shows the output of the SELECT query, displaying 5 rows of employee data:

emp_id	emp_name	salary	hire_date	dept_id
101	Amit	55000.00	2026-01-15	2
102	Neha	65000.00	2025-12-20	2
103	Rahul	45000.00	2026-03-10	1
104	Priya	70000.00	2025-11-18	3
105	Rohan	50000.00	2026-05-01	4

The 'Output' window shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	16:53:47	INSERT INTO Employee VALUES (112,'Anjali',68000,'2026-06-10',1), (113,'Rakesh',72000,'2026-06-12',2), (114,'Meena',48000,'2026-06-15',3), (115,'Suresh',90000,'2026-06-18',4), (116,'Kavita',62000,'2026-06-20',2);	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
2	16:54:11	SELECT * FROM Employee LIMIT 0, 50000	15 row(s) returned	0.000 sec / 0.000 sec

The screenshot shows the MySQL Workbench interface. The 'Query 1' window contains the following SQL code:

```
114 -- (115,'Suresh',90000,'2026-06-18',4),
115 -- (116,'Kavita',62000,'2026-06-20',2);
116 -- SELECT * FROM Employee;
117 • SELECT
118   emp_name,
119   salary,
120   RANK() OVER (ORDER BY salary DESC) AS Salary_Rank
121 FROM Employee;
```

The 'Result Grid' shows the output of the RANK() query, displaying 5 rows of employee data with their salary ranks:

emp_name	salary	Salary_Rank
Suresh	90000.00	1
Sneha	80000.00	2
Pooja	75000.00	3
Rakesh	72000.00	4
Priya	70000.00	5

The 'Output' window shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	16:53:47	INSERT INTO Employee VALUES (112,'Anjali',68000,'2026-06-10',1), (113,'Rakesh',72000,'2026-06-12',2), (114,'Meena',48000,'2026-06-15',3), (115,'Suresh',90000,'2026-06-18',4), (116,'Kavita',62000,'2026-06-20',2);	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
2	16:54:11	SELECT * FROM Employee LIMIT 0, 50000	15 row(s) returned	0.000 sec / 0.000 sec
3	16:56:35	SELECT emp_name, salary, RANK() OVER (ORDER BY salary DESC) AS Salary_Rank FROM Employee;	15 row(s) returned	0.016 sec / 0.000 sec

2: Handle Window Functions

The screenshot shows the MySQL Workbench interface with a query titled "SQL_Internship_Task". The query is as follows:

```
119 -- salary,
120 -- RANK() OVER (ORDER BY salary DESC) AS Salary_Rank
121 -- FROM Employee;
122 • SELECT
123     emp_name,
124     salary,
125     ROW_NUMBER() OVER (ORDER BY salary DESC) AS row_num
126 FROM Employee;
```

The Result Grid shows the following data:

emp_name	salary	row_num
Suresh	90000.00	1
Sneha	80000.00	2
Pooja	75000.00	3
Rakesh	72000.00	4
Priya	70000.00	5

The Output tab shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
2	16:54:11	SELECT * FROM Employee LIMIT 0, 50000	15 row(s) returned	0.000 sec / 0.000 sec
3	16:56:35	SELECT emp_name, salary, RANK() OVER (ORDER BY salary DESC) AS Salary_Rank FROM ...	15 row(s) returned	0.016 sec / 0.000 sec
4	16:57:46	SELECT emp_name, salary, ROW_NUMBER() OVER (ORDER BY salary DESC) AS Row_Num...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to ...	0.015 sec
5	16:58:38	SELECT VERSION() LIMIT 0, 50000	1 row(s) returned	0.000 sec / 0.000 sec
6	16:59:30	SELECT ROW_NUMBER() OVER () AS row_num, emp_name FROM Employee	15 row(s) returned	0.000 sec / 0.000 sec
7	17:00:40	SELECT emp_name, salary, ROW_NUMBER() OVER (ORDER BY salary DESC) AS row...	15 row(s) returned	0.000 sec / 0.000 sec

The screenshot shows the MySQL Workbench interface with a query titled "SQL_Internship_Task". The query is as follows:

```
124 -- salary,
125 -- ROW_NUMBER() OVER (ORDER BY salary DESC) AS row_num
126 -- FROM Employee;
127 • SELECT
128     emp_name,
129     salary,
130     RANK() OVER (ORDER BY salary DESC) AS salary_rank
131 FROM Employee;
```

The Result Grid shows the following data:

emp_name	salary	salary_rank
Suresh	90000.00	1
Sneha	80000.00	2
Pooja	75000.00	3
Rakesh	72000.00	4
Priya	70000.00	5

The Output tab shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
3	16:56:35	SELECT emp_name, salary, RANK() OVER (ORDER BY salary DESC) AS Salary_Rank FROM ...	15 row(s) returned	0.016 sec / 0.000 sec
4	16:57:46	SELECT emp_name, salary, ROW_NUMBER() OVER (ORDER BY salary DESC) AS Row_Num...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to ...	0.015 sec
5	16:58:38	SELECT VERSION() LIMIT 0, 50000	1 row(s) returned	0.000 sec / 0.000 sec
6	16:59:30	SELECT ROW_NUMBER() OVER () AS row_num, emp_name FROM Employee	15 row(s) returned	0.000 sec / 0.000 sec
7	17:00:40	SELECT emp_name, salary, ROW_NUMBER() OVER (ORDER BY salary DESC) AS row...	15 row(s) returned	0.000 sec / 0.000 sec
8	17:05:36	SELECT emp_name, salary, RANK() OVER (ORDER BY salary DESC) AS salary_rank F...	15 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Navigator: Filter objects

SCHEMAS

data_science

Tables

bachchalog

customers

department

dhurandar

emp

employee

excellent

Administration Schemas

Information

No object selected

Query 1: SQL_Internship_Task

```

129 -- salary,
130 -- RANK() OVER (ORDER BY salary DESC) AS salary_rank
131 -- FROM Employee;
132 SELECT
133     emp_name,
134     salary,
135     DENSE_RANK() OVER (ORDER BY salary DESC) AS d_rank
136 FROM Employee;

```

Result Grid

emp_name	salary	d_rank
Suresh	90000.00	1
Sneha	80000.00	2
Pooja	75000.00	3
Rakesh	72000.00	4
Priya	70000.00	5

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	16:58:38	SELECT VERSION() LIMIT 0, 50000	1 row(s) returned	0.000 sec / 0.000 sec
6	16:59:30	SELECT ROW_NUMBER() OVER () AS row_num, emp_name FROM Employee	15 row(s) returned	0.000 sec / 0.000 sec
7	17:00:40	SELECT emp_name, salary, ROW_NUMBER() OVER (ORDER BY salary DESC) AS row_num FROM Employee	15 row(s) returned	0.000 sec / 0.000 sec
8	17:05:36	SELECT emp_name, salary, RANK() OVER (ORDER BY salary DESC) AS salary_rank FROM Employee	15 row(s) returned	0.000 sec / 0.000 sec
9	17:06:42	SELECT emp_name, salary, DENSE_RANK() OVER (ORDER BY salary DESC) AS dens_rank FROM Employee	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to ...	0.000 sec
10	17:07:15	SELECT emp_name, salary, DENSE_RANK() OVER (ORDER BY salary DESC) AS d_rank FROM Employee	15 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

3: Handle CASE WHEN

MySQL Workbench

Navigator: Filter objects

SCHEMAS

data_science

Tables

bachchalog

customers

department

dhurandar

emp

employee

excellent

Administration Schemas

Information

No object selected

Query 1: SQL_Internship_Task

```

137 SELECT
138     emp_name,
139     salary,
140     CASE
141         WHEN salary >= 80000 THEN 'High Salary'
142         WHEN salary >= 60000 THEN 'Medium Salary'
143         ELSE 'Low Salary'
144     END AS salary_category

```

Result Grid

emp_name	salary	salary_category
Amit	55000.00	Low Salary
Neha	65000.00	Medium Salary
Rahul	45000.00	Low Salary
Priya	70000.00	Medium Salary
Rohan	50000.00	Low Salary

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	17:00:40	SELECT emp_name, salary, ROW_NUMBER() OVER (ORDER BY salary DESC) AS row_num FROM Employee	15 row(s) returned	0.000 sec / 0.000 sec
8	17:05:36	SELECT emp_name, salary, RANK() OVER (ORDER BY salary DESC) AS salary_rank FROM Employee	15 row(s) returned	0.000 sec / 0.000 sec
9	17:06:42	SELECT emp_name, salary, DENSE_RANK() OVER (ORDER BY salary DESC) AS dens_rank FROM Employee	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to ...	0.000 sec
10	17:07:15	SELECT emp_name, salary, DENSE_RANK() OVER (ORDER BY salary DESC) AS d_rank FROM Employee	15 row(s) returned	0.000 sec / 0.000 sec
11	17:09:17	SELECT emp_name, salary, CASE WHEN salary >= 80000 THEN 'High Salary' WHEN salary >= 60000 THEN 'Medium Salary' ELSE 'Low Salary' END AS salary_category FROM Employee	15 row(s) returned	0.000 sec / 0.000 sec
12	17:10:07	SELECT emp_name, salary, CASE WHEN salary >= 80000 THEN 'High Salary' WHEN salary >= 60000 THEN 'Medium Salary' ELSE 'Low Salary' END AS salary_category FROM Employee	15 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

4: Views

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'data_science' selected. The main editor shows a SQL query for creating a view named 'Employee_View AS'. The query is as follows:

```
145 -- FROM Employee;
146 CREATE VIEW Employee_View AS
147 SELECT
148     emp_name,
149     salary,
150     hire_date
151 FROM Employee;
152 SELECT * FROM Employee View;
```

The 'Result Grid' tab is active, displaying the data from the 'Employee_View' table. The data is as follows:

emp_name	salary	hire_date
Amit	55000.00	2026-01-15
Neha	65000.00	2025-12-20
Rahul	45000.00	2026-03-10
Priva	70000.00	2025-11-18
Rohan	50000.00	2026-05-01

The 'Output' tab shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
9	17:06:42	SELECT emp_name, salary, DENSE_RANK() OVER (ORDER BY salary DESC) AS dens...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to ...	0.000 sec
10	17:07:15	SELECT emp_name, salary, DENSE_RANK() OVER (ORDER BY salary DESC) AS d_ra...	15 row(s) returned	0.000 sec / 0.000 sec
11	17:09:17	SELECT emp_name, salary, CASE WHEN salary >= 80000 THEN 'High Salary' ...	15 row(s) returned	0.000 sec / 0.000 sec
12	17:10:07	SELECT emp_name, salary, CASE WHEN salary >= 80000 THEN 'High Salary' ...	15 row(s) returned	0.000 sec / 0.000 sec
13	17:13:58	CREATE VIEW Employee_View AS SELECT emp_name, salary, hire_date FROM Emplo...	0 row(s) affected	0.078 sec
14	17:14:39	SELECT * FROM Employee_View LIMIT 0, 50000	15 row(s) returned	0.016 sec / 0.000 sec

5: Triggers

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'data_science' selected. The main editor shows a SQL query for creating a table named 'Employee_Log'. The query is as follows:

```
143 -- ELSE 'Low Salary'
144 -- END AS salary_category
145 -- FROM Employee;
146 -- CREATE VIEW Employee_View AS
147 -- SELECT
148 --     emp_name,
149 --     salary,
150 --     hire_date
151 -- FROM Employee;
152 -- SELECT * FROM Employee_View;
153 CREATE TABLE Employee_Log(
154     log_id INT AUTO_INCREMENT PRIMARY KEY,
155     emp_id INT,
156     action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
157 );
```

The 'Output' tab shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
10	17:07:15	SELECT emp_name, salary, DENSE_RANK() OVER (ORDER BY salary DESC) AS d_ra...	15 row(s) returned	0.000 sec / 0.000 sec
11	17:09:17	SELECT emp_name, salary, CASE WHEN salary >= 80000 THEN 'High Salary' ...	15 row(s) returned	0.000 sec / 0.000 sec
12	17:10:07	SELECT emp_name, salary, CASE WHEN salary >= 80000 THEN 'High Salary' ...	15 row(s) returned	0.000 sec / 0.000 sec
13	17:13:58	CREATE VIEW Employee_View AS SELECT emp_name, salary, hire_date FROM Emplo...	0 row(s) affected	0.078 sec
14	17:14:39	SELECT * FROM Employee_View LIMIT 0, 50000	15 row(s) returned	0.016 sec / 0.000 sec
15	17:16:24	CREATE TABLE Employee_Log(log_id INT AUTO_INCREMENT PRIMARY KEY; emp_id...	0 row(s) affected	0.109 sec

MySQL Workbench

File Edit View Query Database Server Tools Scripting Help

Navigator: junedat x

SCHEMAS

Filter objects

data_science

Tables

bachchalog

customers

department

dhurandar

emp

employee

excellent

Administration Schemas

Information

No object selected

Query 1: SQL_Internship_Task

```

155 -- emp_id INT,
156 -- action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
157 -- );
158 DELIMITER $$
159
160 CREATE TRIGGER Employee_Insert_Log
161 AFTER INSERT
162 ON Employee
163 FOR EACH ROW
164 BEGIN
165     INSERT INTO Employee_Log(emp_id)
166     VALUES(NEW.emp_id);
167 END$$
168
169 DELIMITER ;

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	17:09:17	SELECT emp_name, salary, CASE WHEN salary >= 80000 THEN 'High Salary'	15 row(s) returned	0.000 sec / 0.000 sec
12	17:10:07	SELECT emp_name, salary, CASE WHEN salary >= 80000 THEN 'High Salary'	15 row(s) returned	0.000 sec / 0.000 sec
13	17:13:58	CREATE VIEW Employee_View AS SELECT emp_name, salary, hire_date FROM Emplo...	0 row(s) affected	0.078 sec
14	17:14:39	SELECT * FROM Employee_View LIMIT 0, 50000	15 row(s) returned	0.016 sec / 0.000 sec
15	17:16:24	CREATE TABLE Employee_Log(log_id INT AUTO_INCREMENT PRIMARY KEY, emp_id...	0 row(s) affected	0.109 sec
16	17:17:41	CREATE TRIGGER Employee_Insert_Log AFTER INSERT ON Employee FOR EACH ROW BE...	0 row(s) affected	0.031 sec

Object Info Session

MySQL Workbench

File Edit View Query Database Server Tools Scripting Help

Navigator: junedat x

SCHEMAS

Filter objects

data_science

Tables

bachchalog

customers

department

dhurandar

emp

employee

excellent

Administration Schemas

Information

No object selected

Query 1: SQL_Internship_Task

```

157 -- );
158 -- DELIMITER $$
159
160 -- CREATE TRIGGER Employee_Insert_Log
161 -- AFTER INSERT
162 -- ON Employee
163 -- FOR EACH ROW
164 -- BEGIN
165 --     INSERT INTO Employee_Log(emp_id)
166 --     VALUES(NEW.emp_id);
167 -- END$$
168
169 -- DELIMITER ;
170 INSERT INTO Employee VALUES
171 (117, 'Deepak', 65000, '2026-06-25', 1);

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	17:10:07	SELECT emp_name, salary, CASE WHEN salary >= 80000 THEN 'High Salary'	15 row(s) returned	0.000 sec / 0.000 sec
13	17:13:58	CREATE VIEW Employee_View AS SELECT emp_name, salary, hire_date FROM Emplo...	0 row(s) affected	0.078 sec
14	17:14:39	SELECT * FROM Employee_View LIMIT 0, 50000	15 row(s) returned	0.016 sec / 0.000 sec
15	17:16:24	CREATE TABLE Employee_Log(log_id INT AUTO_INCREMENT PRIMARY KEY, emp_id...	0 row(s) affected	0.109 sec
16	17:17:41	CREATE TRIGGER Employee_Insert_Log AFTER INSERT ON Employee FOR EACH ROW BE...	0 row(s) affected	0.031 sec
17	17:18:36	INSERT INTO Employee VALUES (117,'Deepak',65000,'2026-06-25',1)	1 row(s) affected	0.032 sec

Object Info Session

MySQL Workbench

junedq1 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

data_science

tables

bachchalag

customers

department

dhurandar

emp

employee

excellent

Query 1 SQL_Internship_Task x

Limit to 50000 rows

```
166 -- VALUES(NEW.emp_id);
167 -- END$$
168
169 -- DELIMITER ;
170 -- INSERT INTO Employee VALUES
171 -- (117,'Deepak',65000,'2026-06-25',1);
172 • SELECT * FROM Employee_Log;
```

Result Grid

log_id	emp_id	action_time
1	117	2026-06-25 17:18:36

No object selected

Administration Schemas

Information

Output

Employee_Log 11 x

Apply Context Help Snippets

#	Time	Action	Message	Duration / Fetch
13	17:13:58	CREATE VIEW Employee_View AS SELECT emp_name, salary, hire_date FROM Emplo...	0 row(s) affected	0.078 sec
14	17:14:39	SELECT * FROM Employee_View LIMIT 0, 50000	15 row(s) returned	0.016 sec / 0.000 sec
15	17:16:24	CREATE TABLE Employee_Log(log_id INT AUTO_INCREMENT PRIMARY KEY, emp_id...	0 row(s) affected	0.109 sec
16	17:17:41	CREATE TRIGGER Employee_Insert_Log AFTER INSERT ON Employee FOR EACH ROW BE...	0 row(s) affected	0.031 sec
17	17:18:36	INSERT INTO Employee VALUES (117,'Deepak',65000,'2026-06-25',1)	1 row(s) affected	0.032 sec
18	17:19:31	SELECT * FROM Employee_Log LIMIT 0, 50000	1 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.